

Seeding Monte Carlo algorithms

Gutiérrez-Peña, Eduardo
eduardo@sigma.iimas.unam.mx

Aguirre-Pérez, Román
roman@sigma.iimas.unam.mx

July 9, 2026

Abstract

The primary goal of this work is to illustrate the need of seeding Monte Carlo algorithms when dealing with unobserved variables.

1 Introduction

In real-world scenarios, we often aim to explain a set of traits of a phenomenon of interest under specific conditions. From a statistical perspective, we usually propose a stochastic mechanism to approximate the underlying law that is governing such a phenomenon of interest, say $\pi(\mathbf{z}|\boldsymbol{\theta})$, where $\mathbf{z}$ is a set of attributes and $\boldsymbol{\theta}$ is a random vector that fully defines the proposed stochastic mechanism. Within the Bayesian approach, we further elicitate a propobalistic distribution for $\boldsymbol{\theta}$, say $\pi(\boldsymbol{\theta})$.

2 Literature Review

[Maa et al. \(1996\)](#)

3 Methodology

Monte Carlo methods, such as MCMC and HMC, are widely used in practice for reconstructing the posterior distribution $\pi(\boldsymbol{\theta}|\mathbf{z})$. (Here we explain how they are implemented). A more challenging scenario is when we only

observed a subset of $\mathbf{z}$, that is, if $\mathbf{z} = (\mathbf{x}, \mathbf{y})$ we only have measurements for $\mathbf{y}$, say $\mathbf{y}^{obs}$. In this case, we instead seek to reconstruct $\pi(\mathbf{x}, \boldsymbol{\theta} | \mathbf{y}^{obs})$, and thus, we now also have to initialise $\mathbf{x}$. Leveraging a distance-based approach in tandem with the prior predictive distribution, we proposed the following seeding process for Monte Carlo algorithms when dealing with unobserved variables.

Algorithm 1 summaries such a seeding process.

Algorithm 1 Distance-based Seeding

Require: $\boldsymbol{\theta}^{(t)}$

Ensure: $\boldsymbol{\theta}^{(t)}, \mathbf{x}^{(t)}, \mathbf{y}^{(t)}$

- 1: $t = 0$
 - 2: **repeat**
 - 3: $t \leftarrow t + 1$
 - 4: $\boldsymbol{\theta}^{(t)} \sim \pi(\boldsymbol{\theta})$
 - 5: $\mathbf{x}^{(t)}, \mathbf{y}^{(t)} \sim \pi(\mathbf{x}, \mathbf{y} | \boldsymbol{\theta})$
 - 6: **until** $d(\mathbf{y}^{(t)}, \mathbf{y}^{obs}) < d^{max}$
-

4 Simulations

For a discrete-time setting, let us assume that a population experience an exponential growth with time-varying rate r_t , that is, $N_{t+1} = e^{r_t} N_t$ for $t = 1, 2, \dots, T - 1$. We can rewrite the time-varying rate as $r_t = f_t - m_t$, where $f_t, m_t \in \mathbb{R}^+$ are the fertility and mortality rates at time t , respectively. Now, at each time t , we can think of abundance before and after mortality, say N_t^{pre} and N_t^{post} , respectively. Thus, if M_t denotes the total mortality we have that $N_t^{post} = N_t^{pre} - M_t$.

Given the aforementioned population demographics, we can propose the following stochastic population mechanism:

$$\begin{aligned} M_t &\sim \text{Bin}(N_t^{pre}, 1 - e^{-m_t}) \\ N_{t+1}^{pre} &\sim \text{Poi}(e^{f_t} N_t^{post}) \end{aligned}$$

Furthermore,

$$\begin{aligned}
\log(N_1^{pre}) &\sim \text{Poi}(e^{LN_0}) \\
\log(a_t) &\sim \text{N}(\mu_{la}, \sigma_{la}) \\
\log(b_t) &\sim \text{N}(\mu_{lb}, \sigma_{lb}) \\
\log(f_t) &\sim \text{N}(\mu_{lf}, \sigma_{lf})
\end{aligned}$$

with

$$\begin{aligned}
\mu_{lm} &\sim \text{N}(m_{lm}, sd_{lm}) \\
\sigma_{lm} &\sim \text{N}(0, sd_{lm}) \\
\mu_{lf} &\sim \text{N}(m_{lf}, sd_{lf}) \\
\sigma_{lf} &\sim \text{N}(0, sd_{lf})
\end{aligned}$$

5 Concluding Remarks

References

- Maa, J.-F., Pearl, D. K., and Bartoszyński, R. (1996). Reducing multi-dimensional two-sample data to one-dimensional interpoint comparisons. *The Annals of Statistics*, 24(3):1069–1074.